BharatPay: Predictive Analytics of Digital Payment Transactions

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Abstract— Digital payment platforms generate large volumes of daily transaction data that can reveal adoption trends and support data-driven planning. This paper presents BharatPay, a predictive analytics workflow built using daily Indian payment-system statistics. The study analyses 1,987 daily records covering 1 May 2021 to 8 October 2026 and 15 fields, including transaction counts and values for UPI, IMPS, FASTag, ATM-network and AePS services. Daily UPI transaction count is modelled using calendar variables, historical lags and rolling statistics. Five regression algorithms are evaluated with a chronological 80:20 train–test split. Linear Regression performs best, achieving MAE 13,408,095.84, RMSE 17,904,761.18 and R^2 0.8795. K-Means clustering and PCA are additionally used to identify patterns across payment methods.

Keywords— digital payments, UPI, predictive analytics, time-series features, regression, K-Means, PCA, India.

I. INTRODUCTION

India's digital payment ecosystem includes interoperable systems for person-to-person transfers, merchant payments, toll collection, cash access and recurring payments. Unified Payments Interface (UPI) has become a major channel. Its growth creates an opportunity to study daily transaction behaviour and estimate near-term activity from historical patterns.

Predictive analytics combines historical observations, feature engineering and statistical learning to estimate future outcomes. BharatPay focuses on daily UPI transaction volume and also compares UPI, IMPS, FASTag, ATM-network and AePS activity. Unsupervised clustering is used to identify days with similar payment patterns.

A. Objectives

The objectives are to inspect NPCI statistics; explore UPI transaction count, value and relationships among payment systems; engineer temporal and historical features; compare five regression models against a previous-day baseline; and group daily payment observations using K-Means and PCA.

II. DATASET DESCRIPTION

The dataset contains daily statistics for Indian payment systems. NPCI is the original source of the payment-system statistics; Measure of India provides the compilation and downloadable dataset used in this project. The workbook contains one date field and 14 numeric measures: transaction value and count for seven payment categories.

Attribute	Description
Records / fields	1,987 rows × 15 columns
Date coverage	01-May-2021 to 08-Oct-2026
Target	UPI number of transactions, daily
Payment systems	UPI, IMPS, NACH credit/debit, FASTag, ATM (NFS), AePS
Data quality	0 invalid dates, 0 duplicates, 0 missing numeric values

TABLE I. DATASET SUMMARY

III. DATA PREPROCESSING

The Excel workbook was loaded with pandas. Dates were converted to datetime, records sorted chronologically and duplicate rows checked. Numeric columns were converted to numeric values. The implementation includes interpolation and median filling as safeguards; however, the supplied dataset contained no missing numeric values, so no imputation was required. Rows with invalid dates or negative target values were excluded. The final dataset retained 1,987 observations.

A. Feature Engineering

Five calendar features were generated: year, month, day of week, day of year and weekend indicator. For each of the 14 numeric series, lag features at 1, 7 and 14 days were created, producing 42 historical predictors. Three UPI statistics were calculated using prior observations only: 7-day rolling mean, 14-day rolling mean and 7-day rolling standard deviation. Together these form 50 predictors. The target was day-to-day change in UPI transaction count. After lag and rolling-window rows were removed, 1,973 records remained.

B. Chronological Split

The model-ready observations were kept in date order and divided into the first 80% for training and final 20% for testing. The baseline predicts the current day's count as equal to the previous day's observed count.

IV. EXPLORATORY DATA ANALYSIS

The UPI transaction-count time series shows a sustained upward trajectory over the observation period. Transaction value also rises over time, motivating the use of recent lags and rolling statistics.



Fig. 1. Daily UPI transaction-count trend.

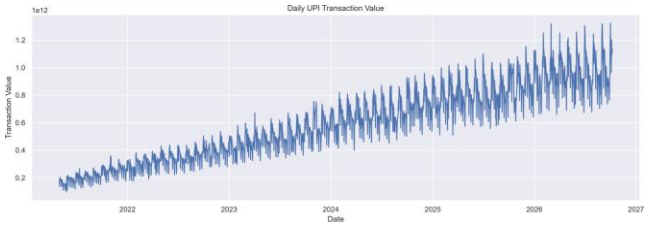


Fig. 2. Daily UPI transaction value.

Average daily transaction counts show UPI as the largest-volume channel in the dataset. Payment services have different operational contexts, so the comparison is descriptive.

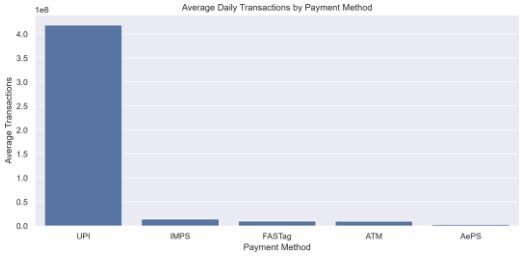


Fig. 3. Average daily transactions by payment method.

The correlation matrix shows positive associations among several value and count series. UPI transaction value and count have a correlation of approximately 0.95. Correlation indicates co-movement, not causation.

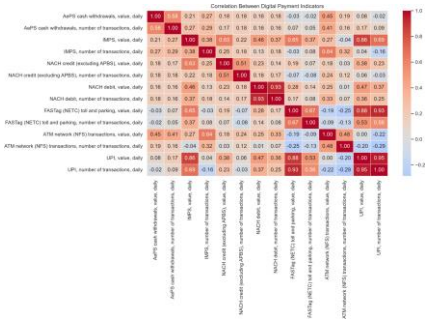


Fig. 4. Correlation heatmap.

V. PREDICTIVE MODELLING METHODOLOGY

The task is supervised regression. Models estimate daily change in UPI count; predicted change is added to the previous day's count to reconstruct the current-day prediction. Negative predictions are clipped to zero.

A. Regression Models

Five scikit-learn regressors were trained. Linear Regression uses ordinary least squares. Decision Tree uses maximum depth 8 and minimum leaf size 5. Random Forest uses 150 trees, maximum depth 12 and minimum leaf size 3. Gradient Boosting uses 150 estimators, learning rate 0.05, depth 3 and Huber loss. SVR uses an RBF kernel with standardisation, C=10 and epsilon=0.05.

Model	Configuration
Linear Regression	Ordinary least squares
Decision Tree	Depth=8; leaf=5
Random Forest	150 trees; depth=12
Gradient Boosting	150; lr=.05; Huber
SVR	Scaled RBF; C=10; ε=.05

TABLE II. MODEL CONFIGURATIONS

B. Evaluation Metrics

MAE measures average absolute error; RMSE penalises larger errors; R^2 measures the fraction of target variance explained. Lower MAE/RMSE and higher R^2 indicate better performance.

C. Unsupervised Clustering

K-Means was applied to UPI, IMPS, FASTag, ATM-network and AePS transaction counts. Variables were transformed using log(1+x) and standardised. Candidate cluster counts from 2 to 6 were evaluated using silhouette score and inertia; the selected solution produced four clusters. PCA reduced the scaled feature space to two components for visualisation.

VI. EXPERIMENTAL RESULTS

A. Regression Performance

Linear Regression achieves the lowest MAE and RMSE and highest R^2. Its R^2 of 0.8795 indicates that approximately 87.95% of variance in the test target is explained. Random Forest ranks second by R^2, followed by SVR and the previous-day baseline.

Model	MAE	RMSE	R^2
Linear Regression	13,408,095.84	17,904,761.18	0.8 95
Random Forest	15,099,042.24	20,181,798.99	0.8 69
SVR	14,595,619.38	20,434,097.90	0.8 30
Previous-day baseline	14,587,220*	20,436,990*	0.8 30
Gradient Boosting	20,236,368.85	25,461,164.68	0.7 63
Decision Tree	21,992,259.25	28,184,899.03	0.7 14

TABLE III. TEST-SET MODEL PERFORMANCE (*rounded console values).

B. Best Model Evaluation

Linear Regression was selected using RMSE, with MAE 13,408,095.84, RMSE 17,904,761.18 and R^2 0.8795. Decision Tree produced the largest errors and lowest R^2.

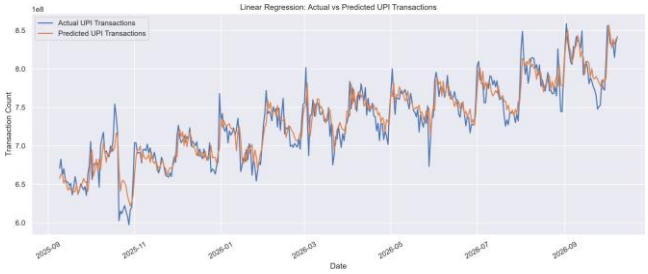


Fig. 5. Actual and predicted UPI counts.

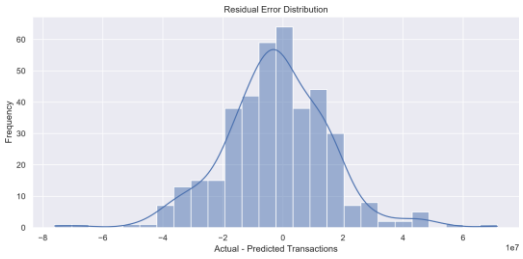


Fig. 6. Residual error distribution.

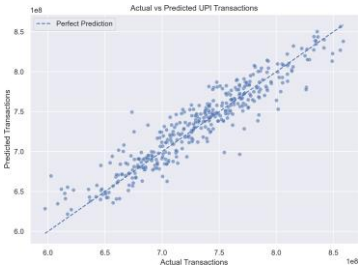


Fig. 7. Actual versus predicted counts.

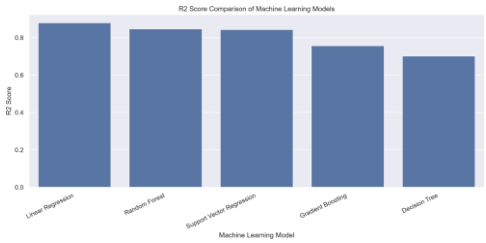


Fig. 8. R^2 comparison.

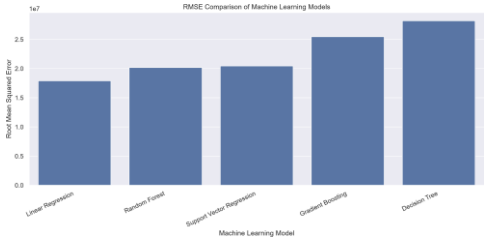


Fig. 9. RMSE comparison.



Fig. 10. MAE comparison.

VII. PAYMENT PATTERN CLUSTERING

Clustering groups days by relative activity across five payment channels. Log transformation reduces scale differences, and standardisation gives each channel a comparable scale. Four clusters were selected. PCA helps visualise the groups, although overlap may remain because PCA preserves overall variance rather than maximising cluster separation.



Fig. 11. K-Means clusters projected using PCA.

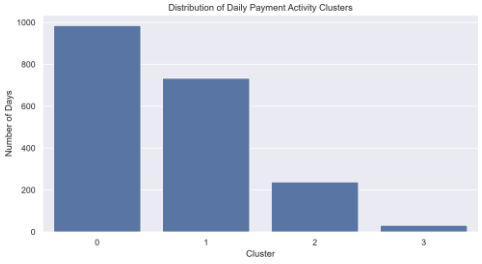


Fig. 12. Distribution across four clusters.

VIII. DISCUSSION AND LIMITATIONS

The results show that engineered historical features provide useful information for estimating daily UPI transaction changes. Linear Regression outperforms the tested tree-based and kernel-based models. The strong previous-day baseline also highlights persistence in transaction volumes. Limitations include the absence of explicit holiday, festival, policy, outage and macroeconomic indicators; evaluation on one final chronological test segment; and the fact that correlation does not establish causation. Cluster interpretation also depends on feature selection and scaling.

IX. CONCLUSION AND FUTURE WORK

BharatPay presents a predictive analytics workflow for Indian digital-payment statistics. The study inspects 1,987 daily records, constructs 50 temporal predictors, compares five regression algorithms and discovers four payment-activity clusters. Linear Regression is the best model on the held-out test set, achieving MAE 13.41 million, RMSE 17.90 million and R^2 0.8795. Future work can incorporate calendar-event indicators, external predictors, rolling-origin validation, prediction intervals and periodic retraining.

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